BIG DATA TOOLS AND PROCEDURES IN POWER SYSTEMS RESEARCH

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October 03, 2017

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OUTLINE

- Big data in power system
- Difficulties dealing with large data
- Big data tools and procedures
- Live demos
  - Folder structure
  - Data Cleaning and Preparation in MS Excel and Notepad++
  - ETL, MS SSDS-BI tools
  - Working table creation, MS SQL Management Studio
  - Analysis, MATLAB

BIG DATA IN POWER SYSTEM

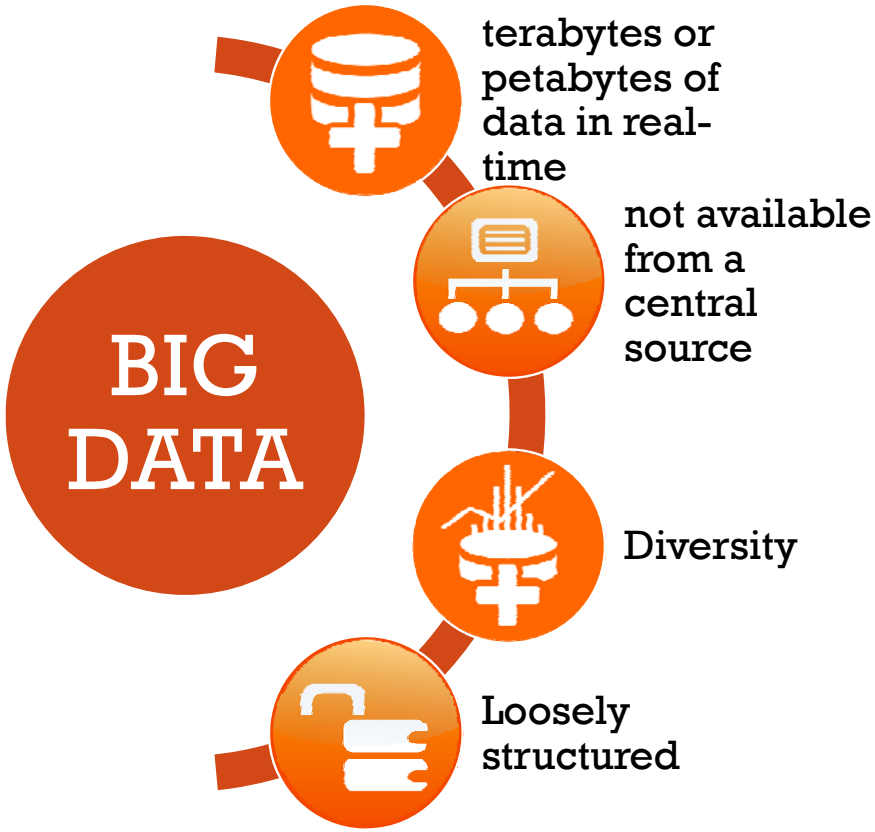
Data Generation in Power System

Past

- SCADA at the transmission level
- Wholesale market operation
- Outage Management System (OMS)

Present

- Advanced Metering Infrastructure (AMI)
- Customer Information System (CIS)
- Geographic Information System (GIS)
- Demand Response Management System (DRMS)
- Internet of things (IoT)
- Home energy management systems (HEMS)
- PtP trading and …



Reading meter once a month ⇨ every 15 minutes ▸ 96 million reads per day for 1 million meters

BIG DATA IN POWER SYSTEM

UQ Gatton PV Plant

- 3.275 MWp PV and 600 kW/760 kWh battery
- 4260 data tags
- Data is stored in delta mode
- Historian Wonderware with relational database (SQL)
- number of tags and the amount of data is growing

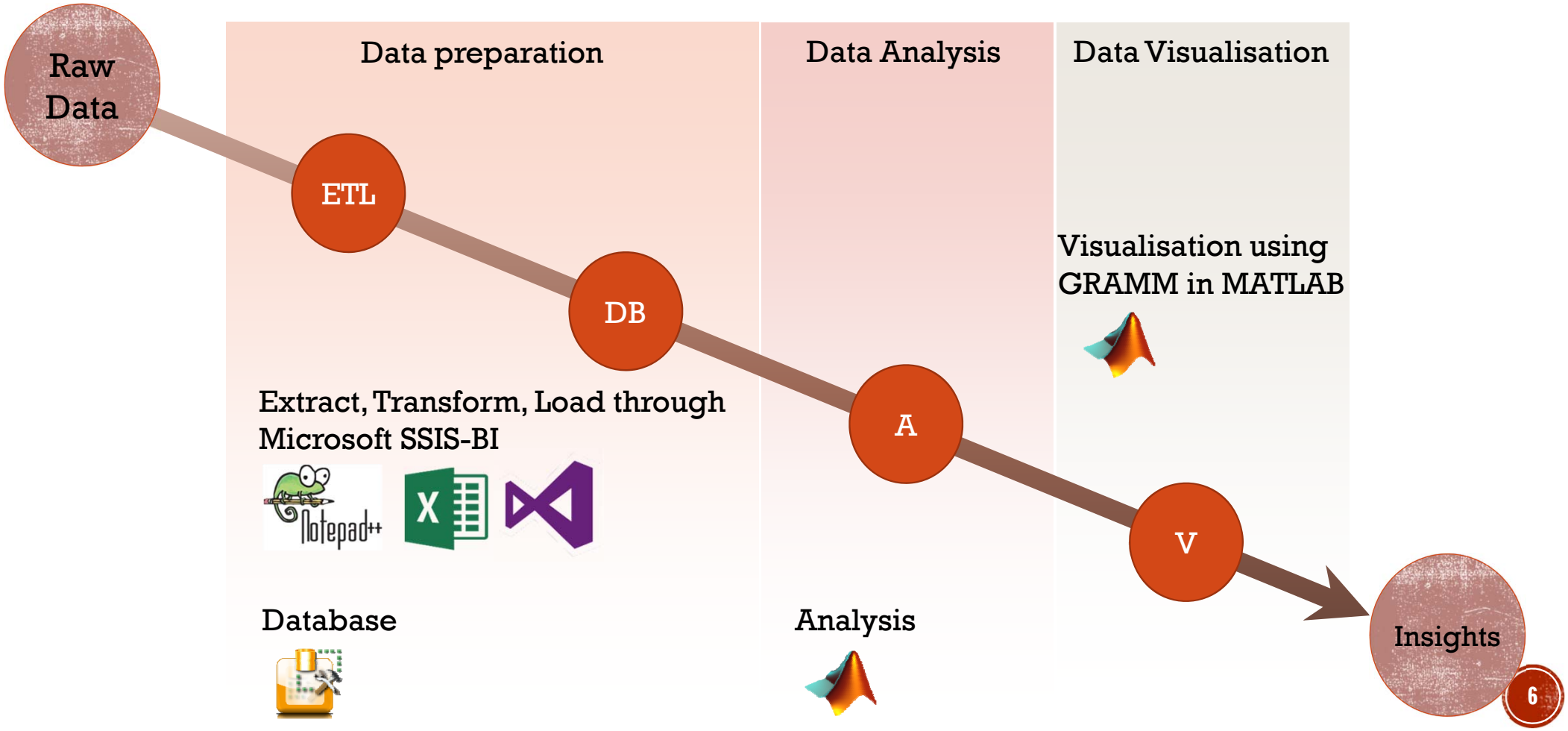
- 4200 points * 20 samples/minute on 64-bit machine ≈ 656 kB/minute ≈ 922 MB/day ≈ **329 GB/year**

- An example: PV ramp-rate on DC and AC sides of PV inverters compared to commands received by battery with date-time tag and battery operation mode flag for a year → 13 * 20 samples/min * 60 * 24 * 365 * 8 ≈ **1 GB**

DIFFICULTIES IN DEALING WITH LARGE DATA

DB: Specifity, combining, automation …

BIG DATA TOOLS & PROCEDURES



LIVE DEMOS

Folder structure

Data Cleaning & Preparation,

MS Excel and Notepad++

ETL,

MS SSIS-BI tools

Working table creation,

MS SQL Management Studio

Analysis,

Calling SQL DB from MATLAB